AI 100 Final Project Report

<u>What I Learned About Gen AI:</u>

During the course of this project, I discovered that generative AI tools like OpenAI GPT models are most useful not simply as answer machines but as a means to guide critical thinking. In the beginning, I expected the AI to directly tell me what was wrong with my code. However, the structure of this assignment forced me to rely on my own reasoning first, which made a difference in how I approached the debugging process.

The most important thing that I noticed was how the AI responded when my self-reflections were weak. Instead of giving me an immediate answer, it provided guidance based on inquiry by asking questions back to me. The questions asked prompted me to analyze my model more thoroughly. I reevaluated assumptions and evidence, and with further reflection about what was actually happening in my model, I was able to reach a rational conclusion. More importantly, this exercise allowed me to understand the logic of that conclusion. For example, when I switched from Logistic Regression to Linear Regression, the results no longer made sense to me. Instead of directly telling me the answer, the AI asked what type of outputs are expected for a classification problem, which helped me realize that Linear Regression predicts continuous values, while my task required class labels like 0 or 1. Another valuable insight was the realization that GenAI is very sensitive to how questions are asked, which I noticed across multiple bug cases in my project. If my reflections were vague or shallow, the assistance I received was also more general. However, when I provided more detailed explanations of what I thought was incorrect, the AI responded with more targeted and helpful hints.

In addition, I learned how GenAI can sometimes give one the sense of being supportive, even when it is not actually solving the problem being asked. There were specific instances when the AI would provide suggestions that sounded reasonable but did not directly fix the issue. I am also aware of the phenomenon known as hallucination, in which AI systems may fabricate information because of its goal of providing a credible answer, sometimes even with a lack of complete knowledge on the subject matter. This was surprising and showed me the importance of evaluating the validity of AI suggestions carefully instead of automatically accepting them as truth.

Overall, this project helped me understand that GenAI is a powerful tool for learning and debugging when used correctly. At this time however, it is a tool with limitations. As advancements in AI systems progress, their ability to process data, make decisions, and simulate human intelligence has unimaginable potential. I found that it is most effective when combined with strong personal reasoning and a clear understanding of the problem. In this way, as AI tools are a guide to problem solving, human interaction is crucial also to guide AI.

What I Learned About Building AI Systems:

This project made me realize how sensitive AI systems can be. Previously, I believed that once a model worked, it would continue to function well even if I changed small elements. Unfortunately, this is not the case. Even small changes can produce a huge impact. For example, removing scaling, switching the model type, or changing how the data is split did not just impact performance slightly. In some cases, it completely broke the model. This helped me visualize how every part of the system matters, and even things that seemed insignificant at first can have immense consequences.

One of the salient takeaways from this project was understanding the importance of data. When I removed scaling, the model struggled much more. I never realized how much machine learning models depend on having inputs that are on similar ranges. This helped me realize that preprocessing is a key part of making a model work.

I was also able to observe that some bugs do not show up as errors, which makes them more difficult to detect. For example, when I shuffled the labels, the code still ran, but the model was no longer learning anything useful. Because of this, I learned the essential need to think about whether the results make logical sense and not just rely on the code running or not. In addition, overfitting of the model stood out for me. Using the wrong evaluation setup or training on the wrong data rendered results misleading. Instead, it made the model less effective with the new data. This is why there is balance when training AI models to perform a task; a greater volume of data is not necessarily better.